

GROUP *Fairness* IN MACHINE LEARNING

What Is Group-Fairness?

Group fairness, also known as demographic parity or disparate impact, aims to ensure that the outcomes of machine learning models are equitable across different demographic groups. It involves assessing and mitigating biases that may lead to unfair treatment or discrimination.



Indpendance

It's goal is to get rid of any connection or correlation between sensitive attributes and characteristics (gender, ethnic background, religious/political affiliation) and the algorithm's predictions. this metric's goal is to make sure there is equal representation within the data given.

Separation

It makes sure that the algorithm's predictive accuracy does not change between groups for those that have attributes in common.



This standard supports the idea of equal opportunity and is necessary for applications such as the hiring system and loan approving systems.



Sufficiency

It makes sure that for all the groups of people recieving the same prediction, the accuracy of these predictions are consistent, regardless of the group affiliation.

